

TradeCircle

Platform Documentation

Deployment Guide & Administrator Handbook

Version 1.0 — 2026

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TradeCircle — Deployment Guide

A complete, beginner-friendly, step-by-step guide to deploying the **TradeCircle** platform to **Vercel** with **Firebase** as the backend and **Cloudinary** for image hosting.

> If this is your first time deploying a Next.js app, follow each section in order. No step is optional unless explicitly marked.

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1. Prerequisites

Before you begin, make sure you have:

Requirement	Purpose	Link
Node.js 20+	Runs the dev server and builds	https://nodejs.org/
Git	Version control	https://git-scm.com/
GitHub account	Hosts your code	https://github.com/
Vercel account (free tier OK)	Hosts the deployed app	https://vercel.com/signup
Firebase account (free Spark plan OK)	Auth, Firestore database, Storage, FCM push	https://firebase.google.com/
Cloudinary account (free tier OK)	Image uploads and CDN	https://cloudinary.com/users/register_free

Verify Node and Git are installed:

```
node --version    # should print v20.x.x or higher
```

```
git --version    # should print git version 2.x.x
```

2. GitHub Setup

If your project is not already in a Git repository:

```
# From the project root
git init
git add .
git commit -m "Initial commit"
```

Create a new **empty** repository on GitHub (do NOT add a README, .gitignore or license — your local repo already has them). Then:

```
git branch -M main
git remote add origin git@github.com:YOUR_USERNAME/tradecircle.git
git push -u origin main
```

> Tip: If you use HTTPS instead of SSH, use

https://github.com/YOUR_USERNAME/tradecircle.git.

3. Firebase Project Setup

3.1 Create the Firebase project

1. Go to <https://console.firebase.google.com/>
2. Click **Add project**, give it a name (e.g. `tradecircle-prod`), accept terms, and (optionally) disable Google Analytics for simpler setup.
3. Wait for the project to be created, then click **Continue**.

3.2 Register a Web App

1. On the project home page click the `</>` Web icon.
2. App nickname: `TradeCircle Web`. **Do NOT** check "Also set up Firebase Hosting".
3. Click **Register app**. Firebase will display a config object — keep this tab open, you'll need these values in Section 5:

```
const firebaseConfig = {
  apiKey: "AIza...",
  authDomain: "tradecircle-prod.firebaseio.com",
  projectId: "tradecircle-prod",
  storageBucket: "tradecircle-prod.firebaseio.com",
  messagingSenderId: "123456789",
  appId: "1:123456789:web:abc123"
};
```

3.3 Enable Authentication

1. In the left sidebar: **Build** → **Authentication** → **Get started**.
2. On the **Sign-in method** tab, enable:
 - **Email/Password** (Save)
 - **Google** — set a project support email, then Save
 - **Phone** (Save) — note: requires billing to be on Blaze plan for production SMS; OK to skip for testing
3. On the **Settings** → **Authorized domains** tab, the default `localhost` and `*.firebaseapp.com` entries are pre-listed. You'll add your custom domain later (Section 8).

3.4 Enable Cloud Firestore

1. Left sidebar: **Build** → **Firestore Database** → **Create database**.
 2. Choose a region close to your users (e.g. `australia-southeast1` for AU/NZ, `us-central1` for North America).
 3. Start in **production mode** (we will apply security rules next).
 4. Once created, go to the **Rules** tab. Open the file `firestore.rules` from this repo, copy its entire contents, paste them into the Firebase console editor, and click **Publish**.
- > Alternative: install the Firebase CLI (`npm i -g firebase-tools`), run `firebase login`, `firebase init firestore`, and then `firebase deploy --only firestore:rules`.

3.5 Enable Cloud Storage

1. Left sidebar: **Build** → **Storage** → **Get started**.
2. Accept the default location (must match your Firestore region in most cases).
3. Start in **production mode**.
4. (Optional) Update Storage rules to mirror Firestore access patterns if you plan to upload files directly to Firebase Storage. The default app uses Cloudinary for images, so this is rarely needed.

3.6 Generate a Service Account (for server-side admin SDK)

1. In the Firebase console, click the **gear icon** → **Project settings** → **Service accounts** tab.
2. Click **Generate new private key**, confirm **Generate key**. A JSON file will download.
3. Open the JSON in a text editor. You'll paste its full contents (as a single line) into the `FIREBASE_SERVICE_ACCOUNT_JSON` env var in Section 5.

> **NEVER** commit this JSON to Git. Treat it like a password.

3.7 Configure Web Push (VAPID key) — optional but recommended

1. **Project settings** → **Cloud Messaging** tab.


```
# ████████████████████████████████████████████████████████████████  
# Firebase Admin (REQUIRED for /api/session, /api/admin/*)  
# Paste the entire service account JSON as a single line  
# ████████████████████████████████████████████████████████████████  
FIREBASE_SERVICE_ACCOUNT_JSON={"type":"service_account","project_id": "...", "private_key": "-----BEGIN PRIVATE KEY-----"  
# ████████████████████████████████████████████████████████████████  
# Cloudinary (REQUIRED for product/avatar image uploads)  
# ████████████████████████████████████████████████████████████████  
NEXT_PUBLIC_CLOUDINARY_CLOUD_NAME=mycloud  
NEXT_PUBLIC_CLOUDINARY_UPLOAD_PRESET=tradecircle_unsigned  
  
# ████████████████████████████████████████████████████████████████  
# Optional integrations  
# ████████████████████████████████████████████████████████████████  
  
# OpenAI – for AI-driven fraud detection, recommendations, etc.  
# Prefer setting via Admin Panel → AI Settings instead of env.  
OPENAI_API_KEY=sk-...  
  
# EmailJS – transactional emails  
EMAILJS_PUBLIC_KEY=  
EMAILJS_SERVICE_ID=  
  
# Stripe – payments  
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=pk_live_...  
STRIPE_PAYMENT_LINK_URL=https://buy.stripe.com/...  
  
# eWAY – alternative payment gateway  
EWAY_API_KEY=  
EWAY_PASSWORD=  
  
# AWS S3 – alternative storage backend (instead of Cloudinary)  
AWS_ACCESS_KEY_ID=  
AWS_SECRET_ACCESS_KEY=  
AWS_REGION=us-east-1  
AWS_S3_BUCKET=  
  
# Google Analytics 4  
NEXT_PUBLIC_GA_MEASUREMENT_ID=G-XXXXXXXXXX
```

> **Tip — pasting the service account JSON into Vercel:** When pasting into Vercel's env var UI, you can paste the multi-line JSON directly into the value field — Vercel preserves it correctly. For your local `.env.local`, you must keep it on a single line (escape newlines in `private_key` are already `\n` in the JSON).

6. Local Development

```
# 1. Install dependencies
npm install

# 2. Create your local env file
cp .env.example .env.local

# 3. Open .env.local and paste the values from Section 5

# 4. Start the dev server
npm run dev
```

Open `http://localhost:3000` — the homepage should load.

First-run: Create the master admin

Visit `http://localhost:3000/setup` in your browser.

- Fill in your full name, email, and a strong password (min 8 chars, 1 uppercase, 1 number).
- Click **Create Master Admin**.
- You'll be redirected to `/admin/dashboard`.

This page locks itself after the first super-admin is created — additional admins must be added from the admin panel (see `ADMIN_GUIDE.md`).

7. Vercel Deployment

7.1 Import the project

1. Go to `https://vercel.com/new`.
2. Click **Import Git Repository** and select your TradeCircle repo.
3. Vercel auto-detects **Next.js**. Leave defaults:

- **Framework Preset:** Next.js
- **Build Command:** `npm run build`
- **Output Directory:** `.next`
- **Install Command:** `npm install`

7.2 Add environment variables

Click **Environment Variables** and add every variable from Section 5.

For each variable:

- **Name:** e.g. `NEXT_PUBLIC_FIREBASE_API_KEY`
- **Value:** paste the actual value
- **Environments:** tick **Production**, **Preview**, and **Development**

> Repeat for all variables. Don't deploy until they're all in.

7.3 Deploy

Click **Deploy**. The first build takes 2-5 minutes. When it finishes you'll get a URL like `https://tradecircle-xyz.vercel.app`.

Visit the URL — the homepage should load. If it doesn't, see Section 10.

8. Custom Domain

8.1 Add domain in Vercel

1. In your Vercel project: **Settings** → **Domains**.
2. Type your domain (e.g. `tradecircle.com.au`), click **Add**.
3. Vercel will show DNS records to configure.

8.2 Configure DNS

Go to your domain registrar (GoDaddy, Cloudflare, Namecheap, etc.) and add:

- **Root domain** (`tradecircle.com.au`): A record → `76.76.21.21`
- **www subdomain**: CNAME → `cname.vercel-dns.com`

Save. DNS propagation usually takes 1-30 minutes.

8.3 Wait for SSL

Vercel auto-provisions a Let's Encrypt SSL cert (typically <1 minute after DNS resolves). The domain badge will turn green.

8.4 Update Firebase authorized domains

1. Firebase console → **Authentication** → **Settings** → **Authorized domains**.
2. Click **Add domain**, enter `tradecircle.com.au` (and `www.tradecircle.com.au` if used).

Without this, Google Sign-In and password reset emails will fail on your production domain.

9. Post-Deployment Verification

Run through this checklist:

- ☐ Homepage loads at your domain
- ☐ `/signup` creates a new user (verify it appears in Firebase → Authentication → Users)
- ☐ `/login` signs in with that user
- ☐ `/admin/dashboard` is accessible after logging in as the admin from Section 6

- `[] /products/new` — create a product, upload an image (verify it appears in Cloudinary dashboard → Media Library)
- `[] /messages` — send a message to another user; verify it appears in real-time
- `[] /settings` — push permission prompt appears (if VAPID is configured)
- `[]` No "Missing Firebase env var" warnings in browser console
- `[]` Vercel → Deployments → latest → Functions tab: no 500 errors in `/api/session` or `/api/admin/*`

10. Troubleshooting

"Missing Firebase env var" warning in console

Some `NEXT_PUBLIC_FIREBASE_*` variable is missing from Vercel.

Fix: Vercel → Settings → Environment Variables — verify all 6 (plus VAPID) are present for the Production environment. Redeploy.

500 error on `/api/session`` or `/api/admin/*``

`FIREBASE_SERVICE_ACCOUNT_JSON` is missing or malformed.

Fix: Re-download the service account from Firebase Project Settings → Service Accounts, paste the entire JSON contents into the Vercel env var (no surrounding quotes needed), redeploy.

Images fail to upload from `/products/new``

Cloudinary upload preset is **signed**, not unsigned.

Fix: Cloudinary → Settings → Upload → Upload presets → edit your preset → Signing mode = **Unsigned** → Save.

Firestore "permission denied" errors in console

Security rules not deployed.

Fix: Either copy `firestore.rules` into Firebase Console → Firestore → Rules and click Publish, or run `firebase deploy --only firestore:rules` from your local repo.

Build fails on Vercel with TypeScript errors

The project ships with `typescript.ignoreBuildErrors: true` already configured in `next.config.ts` to handle harmless type warnings. If you still see errors:

Fix: open `next.config.ts` and confirm:

```
typescript: { ignoreBuildErrors: true },
eslint: { ignoreDuringBuilds: true },
```

Google Sign-In returns "auth/unauthorized-domain"

Your production domain is not in Firebase's authorized domain list.

Fix: Firebase → Authentication → Settings → Authorized domains → Add domain.

Phone auth doesn't work

Phone auth requires the Firebase **Blaze (pay-as-you-go)** plan in production.

Fix: upgrade in Firebase Console → Usage & billing → Modify plan.

Push notifications never arrive

- VAPID key missing or wrong.
- Service worker not registered (check DevTools → Application → Service Workers).

Fix: confirm `NEXT_PUBLIC_FIREBASE_VAPID_KEY` is set, hard-refresh, accept the browser permission prompt.

"Quota exceeded" on Firestore reads

You've hit the free Spark plan's daily quota (50K reads).

Fix: upgrade to Blaze plan (still effectively free for small apps — you pay only for usage above the free allowance).

Need more help? Check the project README for support links, or open an issue on GitHub.

TradeCircle — Admin Guide

A friendly guide to administering your TradeCircle platform. Written for non-technical operators — no coding required.

> **Looking to deploy the app?** See `DEPLOYMENT.md` instead.

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1. First-Time Setup

Right after the app is deployed for the first time, you need to create the master administrator account.

1. Open your domain in a browser: `https://yourdomain.com/setup`

2. Fill in the form:

- **Full Name** — e.g. *Jane Doe*
- **Email** — your admin email (e.g. *admin@yourdomain.com*)
- **Password** — minimum 8 characters, at least 1 uppercase letter and 1 number
- **Confirm Password** — same as above

3. Click **Create Master Admin**.

4. You'll be redirected to `/admin/dashboard`.

> The `/setup` page **locks itself permanently** after the first super-admin is created. All additional admins must be added from inside the admin panel (see Section 4).

2. Logging In

1. Visit `https://yourdomain.com/login`.
2. Enter your admin email and password.
3. After signing in, click your avatar (top right). If you're an admin, you'll see an **Admin Dashboard** link.
4. Click it to enter `/admin/dashboard`.

> If you forget your password, use the **Forgot password?** link on the login page. A reset email will be sent.

3. Admin Dashboard Overview

The admin panel is organized as a left-hand sidebar. Each entry below opens a dedicated page.

Section	What it does
Dashboard	High-level stats: total users, active listings, GMV, recent activity feed.
Users	Manage buyers, sellers and advisors — search, view, ban, unban, delete.
Admin Users	Manage administrator accounts — create new admins, assign roles, toggle module permissions.
Fraud Flags	Review listings/users flagged by AI for suspicious activity. Dismiss or take action.
Products	Moderate marketplace listings — feature, hide, remove, bulk-edit.
Advisories	Moderate advisor posts and articles.
Enquiries	View incoming contact-form submissions and respond.
Orders	Order history, payment status, refunds, fulfillment notes.
Ads	Manage banner ads and sponsored content placement.
Ad Campaigns	A/B test ad creatives and track conversions.
Marketing SSO	Connect Mailchimp / HubSpot / Meta accounts for marketing automation.
Loyalty Program	Configure point earning rules, tiers, and reward catalogue.
AI Settings	Configure OpenAI (or alternative) API key, model, and feature toggles.
CMS	Edit blog posts, FAQ entries, and static pages (About, Terms, Privacy).
Tax	Define regional tax rates (GST/VAT/sales tax).
Feature Toggles	Turn entire modules (chat, advisors, loyalty, etc.) ON/OFF without redeploying.
Configuration	Branding (logo/colors), backend selection (Firebase vs. Supabase), payment gateways, currency, theme.
Tenant Branding	White-label settings for multi-tenant deployments.
Operator Billing	Manage subscription tiers for tenants/operators.
Operators	List and manage multi-tenant operator accounts.
Storage	Switch active storage backend between Cloudinary and AWS S3.
Analytics	KPIs, charts, top sellers, GMV trends.
Integrations	GA4 measurement ID, Facebook Pixel ID, other tracking.
Exports	Download CSV reports (users, orders, products, transactions).
System Backup	Create, restore, and schedule Firestore backups.
Roles	Define regional admin hierarchies (country/region managers).
API Keys	Generate and revoke API keys for third-party integrations.

Section	What it does
System Status	Service Level Agreement (SLA), incident log, current health.
Geo-Targeted Ads	Location-based ad delivery rules.
Data Residency	Choose Firebase region for compliance (e.g. EU, AU).

4. Creating Additional Admins

1. From the sidebar, click **Admin Users**.

2. Click **Add Admin** (top right).

3. Fill in:

- **Name** — the new admin's full name
- **Email** — their work email (login email)
- **Role** — pick one of:
 - **super-admin** — full access to everything
 - **admin** — broad access, but cannot manage other admins
 - **moderator** — limited to user/content moderation
 - **analyst** — read-only access to analytics & exports
- 4. Toggle individual **module permissions** on/off as needed.
- 5. Click **Save**.

The new admin will receive a **password reset email**. They click the link, set their own password, and can then log in at `/login`.

> You can change roles or permissions any time by clicking the admin in the list and editing.

5. User Management

1. Sidebar → **Users**.

2. Use the search bar at the top — search by email, name, or user ID.

3. Click a row to expand the user's detail panel.

4. Available actions:

- **Ban** — sets `active = false`. User cannot log in. Listings hidden.
- **Unban** — restores access.
- **Delete** — permanent. Wipes user record and revokes auth.
- **View Activity** — orders, messages, listings, login history.

> Banning is recoverable. Deletion is **not**. When in doubt, ban first.

6. Authentication & Session

The platform uses HTTP-only secure cookies to maintain logged-in sessions:

Cookie	Purpose
tc-session	Encrypted ID token (validates user identity on every request)
tc-role	User role (buyer/seller/advisor/admin) for fast UI rendering
tc-admin	Admin flag for gated admin routes

- **Session duration:** 7 days. After that, the user must log in again.
- **Logout:** clicking Logout clears all three cookies and revokes the refresh token on Firebase, immediately ending the session on all devices.

7. Permissions System

Each admin has **13 module-level permission flags**. They control which sidebar entries are visible to that admin.

- **super-admin** — all 13 flags automatically `true`. Cannot be reduced. Full power.
- **admin** — preset with most flags `true` except admin-user-management.
- **moderator** — preset focused on Users, Products, Advisories, Fraud Flags, Enquiries.
- **analyst** — preset focused on Analytics, Exports (read-only).

You can override any preset by toggling individual permissions in the Admin Users edit dialog. Saving instantly updates that admin's access (they may need to refresh).

8. Common Workflows

Workflow 1 — Approve or reject a flagged product

1. Sidebar → **Fraud Flags**.
2. Browse the list (most recent first).
3. Click a row to open details — see the flag reason, AI confidence score, and the listing preview.
4. Decide:

- **Dismiss flag** — listing stays live, flag cleared.
- **Remove listing** — listing is hidden, seller notified.
- **Ban seller** — escalate if the seller has multiple flags.

Workflow 2 — Ban a problem user

1. Sidebar → **Users**.
2. Search by email or name.
3. Click their row.
4. Click **Ban** → confirm. The user is signed out immediately on all devices.

Workflow 3 — Update site branding

1. Sidebar → **Configuration**.
2. Open the **Branding** tab.
3. Upload your logo (PNG or SVG), set primary/accent colors, set tagline.
4. Click **Save**. Changes go live within ~30 seconds.

Workflow 4 — Schedule a weekly Firestore backup

1. Sidebar → **System Backup**.
2. Open the **Auto Backup** tab.
3. Toggle **Enabled** on.
4. Frequency: **Weekly**, Day: **Sunday**, Time: **02:00**.
5. Retention: 8 weeks.
6. Click **Save**. The system will run the backup automatically.

Workflow 5 — Set up Google Analytics 4 tracking

1. Get your GA4 Measurement ID from <https://analytics.google.com/> (format: G-XXXXXXXXXX).
2. Sidebar → **Integrations**.
3. Paste the ID into the **GA4 Measurement ID** field.
4. Click **Save**. Pageviews start tracking within a few minutes.

9. Maintenance Schedule

A recommended cadence to keep the platform healthy:

Frequency	Tasks
Daily	Review Fraud Flags , respond to open Enquiries , scan System Status for incidents.
Weekly	Review Analytics trends, verify last Backup ran, check Orders for stuck payments.
Monthly	Review user growth, subscription/loyalty metrics, top sellers, and revenue.
Quarterly	Clean up archived data, rotate API keys , review admin user list (remove ex-employees), audit role permissions.

10. Troubleshooting

"Can't log in"

- Use **Forgot password?** on the login page to reset.
- Make sure the email is exactly the one you registered with.
- If still stuck, ask a super-admin to delete and re-create your admin record.

"Permission denied" when opening a sidebar item

- Your admin role doesn't include that module flag.
- Ask a super-admin to enable the module for you in **Admin Users** → your row → toggle the flag.

"Page won't load" / blank screen

- Check **System Status** for active incidents.
- Hard refresh the browser (Ctrl/Cmd + Shift + R).
- Try an Incognito window to rule out a broken cookie.
- Check the deployed app's status page in Vercel.

"Image won't upload"

- Verify in **Storage** that an upload backend is selected and credentials are filled in.
- Ensure your Cloudinary preset is **unsigned** (see `DEPLOYMENT.md` §4.3).

"AI features aren't working"

- Sidebar → **AI Settings** — verify the OpenAI API key is set and the toggle is on.
- Check that your OpenAI account has credit.

"User got an email I didn't intend to send"

- Sidebar → **CMS / Marketing SSO** — review automation rules.
- Disable any automation triggers you don't recognize.

Need more help? Contact your technical administrator or refer to `DEPLOYMENT.md` for infrastructure-level issues.